

LEO ZHOU

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Education

Rice University | B.S. in Computer Science | **GPA: 3.92/4.00** **Expected Graduation May 2028**

- **Coursework:** Data Structures & Algorithms, Linear Algebra, Systems Programming, Intro to Programming, Intro to Computer Engineering, Multivariable Calculus, Discrete Math & Graph Theory, Digital Logic Design

Work Experience

Software Engineer | **KEWW** **September 2026 – Present**

- Building Cashew, a Next.js/TypeScript Caterease replacement deployed on AWS and used by 12 employees
- Integrating the Gemini API to turn notes into proposals, BEOs, invoices and tasting documents for review
- Indexing 200,000 email threads and 15 years of records into Postgres with pgvector, PII redacted by Presidio
- Developing SEO and marketing agents that draft reviewed web pages and campaigns from approved sources

Software Developer | **RiceApps** | Rice Football Project **September 2026 – Present**

- Building Rice Football's data backend to bring vendor feeds into a shared database with each source tracked
- Planning R/Python data access and cloud report execution so coaches can request fresh reports on demand

Undergraduate Research Assistant | **Rice University** | Prof. Jiarong Xing **August 2026 – Present**

- Investigating how collaborating AI agents can exceed access policies for a systems-security research proposal
- Fixed daily re-logins and missing remote usage in a Swift macOS app with OAuth refresh and SSH transcript sync
- Eliminated repeat Keychain permission prompts by replacing ad-hoc app signing with a stable developer identity

Software Engineer Intern | **CanSemi** | Test Engineering Team **June 2026 – August 2026**

- Parsed 500+ binary test equipment logs in Python into a normalized PostgreSQL schema, 40 lots in under 2 min
- Wrote a FastAPI service with 8 endpoints over that data, cutting report turnaround ~75% for 6 engineers
- Built an interactive React and Canvas wafer map with hover and drill-down, retiring the team's Excel heatmaps

Software Engineer Intern | **Inkstone Technologies** **April 2026 – June 2026**

- Built a SwiftUI reading app with Apple Vision OCR to add pronunciation only for unfamiliar Chinese characters
- Parsed and frequency-ranked a 113,000-entry Chinese dictionary in Python into an indexed SQLite database

Projects

BlunderNet [🐙](#) | [blundernet.com](#) [🌐](#) | *Go, React, PostgreSQL, Redis, AWS, Terraform, Docker* **Ongoing**

- Built a chess training site for ~200 users, with 3.25M puzzles filtered by rating, theme and game phase
- Cut uniform random puzzle selection from 1.4 s to 0.9 ms with a precomputed grid and a stored shuffle key
- Defined a separate AWS deployment in Terraform with a stateless Go API, Redis state and SQS engine workers

Vestigo [🐙](#) | [vestigo.earth](#) [🌐](#) | *Python, PyTorch, CLIP/SigLIP, tool calling, calibration* **Ongoing**

- Building a geolocation agent with citations. Baseline 88-90% correct at claimed granularity on 28 photos

From Scratch [🐙](#) | HackRice 16 | [fromscratch](#) [🌐](#) | *TypeScript, React, Tailwind, Gemini API* **September 2026**

- Built a crafting game that teaches how everyday objects are made and the materials and labour behind them

Leadership & Activities

Rice Fencing Club | Co-president, Coach & Website Manager | Three SWIFA wins **August 2025 – Present**

- Built and manage [ricefence.com](#) [🌐](#) with scroll-driven animations for recruitment, photos and club info

Junior National Fencing Team | Captain, Coach & Referee | Junior Worlds + Commonwealth **2022 – 2025**

- Won 30+ titles across Asia-Pacific, coached 100+ fencers and refereed 100+ bouts

Physics Pathfinders | Charity President | Physics workshops for 600+ students across 22 schools **2023 – 2025**

Technical Skills

Languages: Python, Go, TypeScript/JavaScript, SQL, Swift, C/C++, HTML/CSS

Frameworks & Data: React, Next.js, FastAPI, SwiftUI, PyTorch, PostgreSQL, Redis, SQLite, pgvector

Systems & Tools: AWS (ECS, SQS, RDS), Terraform, Docker, Linux, Git, GitHub Actions, WebSockets, ESP32